

Introduction to Data Mining

Procheta Sen



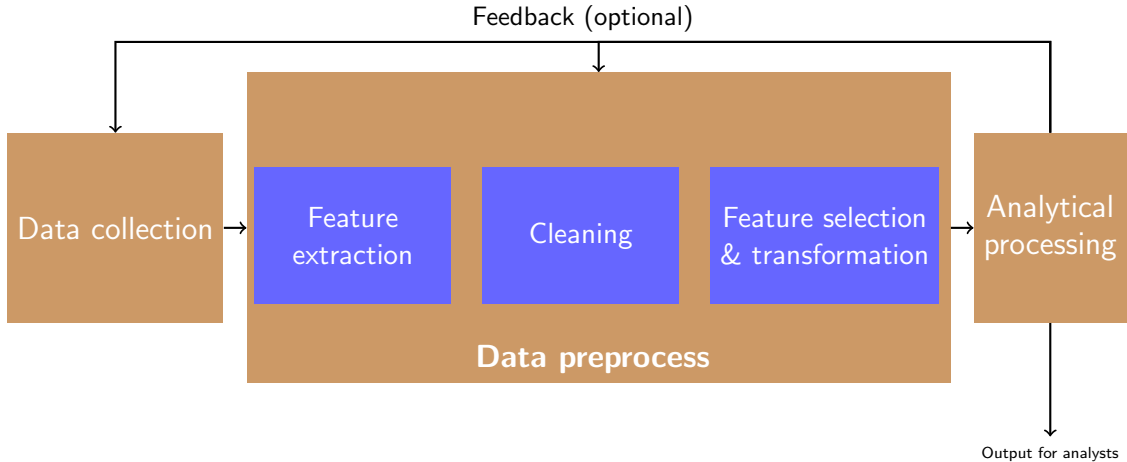
What is Data Mining?

Automated analysis of massive data.

Why Data Mining?

- ▶ The exponential growth of data from terabytes to petabytes.
- ▶ Major sources of abundant data (e.g. social media, YouTube).
- ▶ Data collection has become easier.

The Data Mining Process



Data Collection Phase

- ▶ Highly application-specific. For example:
 - ▶ Sensor networks
 - ▶ User surveys
 - ▶ Automatically collected documents
- ▶ **Critically important:** may significantly impact the whole data mining process
- ▶ Once data is collected it can be stored in a structured or flat format.

Data Preprocessing Phase

Feature extraction

- ▶ convert data into a format that is friendly to data mining algorithms

Data cleaning

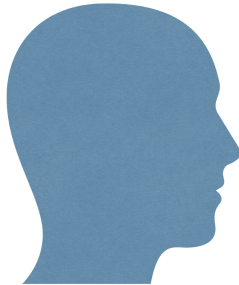
- ▶ missing and erroneous parts of the data are either estimated or corrected

Feature selection and transformation

- ▶ remove irrelevant features
- ▶ transform existing features to features of different scale or format

Feature Extraction

Raw data



analyst abstract out
the features that are
most relevant to
a particular application

Data into a
data-mining-algorithms
friendly format, such as

- **Multidimensional**
- Time series
- Semistructured format
- ...

Multidimensional format
every data point (object) is characterised
by a sequence of fields
(**features**, **attributes**, or **dimensions**)

Features

- ▶ Features are central to data mining
- ▶ Features allow us to abstract data points and learn rules that can be used to predict things for unseen/future data points
 - ▶ If we learn the rule that

`if colour == red then flower = rose`

- ▶ we can use this rule to classify not only flowers in our train dataset, but also all flowers including ones that we do not have in our train dataset
- ▶ Coming up with good features is an art!
 - ▶ feature-engineering (https://en.wikipedia.org/wiki/Feature_engineering)
- ▶ Recent work in machine learning such as deep learning focus on automatically discovering good features from data, without any human intervention

Objects and Features

ID	Height (cm)	Weight (kg)	Marital status	Employed
1	175	60	Single	Yes
2	168	80	Married	Yes
3	183	85	Single	No
4	178	65	Divorced	Yes
5	194	90	Married	No
6	185	78	Married	Yes

A feature is

a property or a characteristic of an object

An object is

described by a collection of attributes

Data Cleaning

- ▶ Extracted data may have erroneous or missing fields
- ▶ Possible actions:
 - ▶ drop a record
 - ▶ estimate a missing value
 - ▶ remove inconsistencies

Feature Selection & Transformation

- ▶ Choosing meaningful and effective features for data mining algorithms are important
- ▶ Possible actions:
 - ▶ identify and remove irrelevant features
 - ▶ transform the current set of features to a new data space
- ▶ Data transformation
 - ▶ Transform attributes to new attributes (e.g., numerical age $\rightarrow$ {young, middle-aged, elderly})

Feature Pruning

- ▶ There might be many **irrelevant** features (i.e. features that are completely uncorrelated with the prediction)
- ▶ Can an algorithm at hand deal well with **redundant features**?
- ▶ Throw away irrelevant feature?
 - ▶ Irrelevant features for text data: a word appear either almost always or almost never
 - ▶ Irrelevant features for numerical data: low variance features

Example

A retailer has Web logs of customer accesses to Webpages of their website.

Each of the Webpages corresponds to a product.

The retailer stores demographic information for each customer.

Task

design system of targeted product recommendations to customers using the customers demographics and buying behaviour

Example Process

Example

Extract records from a log file

Log file

```
98.206.207.157 - - [31/Jul/
2013:18:09:38 -0700] "GET /
productA.htm
HTTP/1.1" 200 328177 "-" "Mozilla/
5.0 (Mac OS X) AppleWebKit/536.26
(KHTML, like Gecko) Version/6.0
Mobile/10B329 Safari/8536.25"
"retailer.net"
```

Create one record for each customer: Number of accesses to each product description

Remove log records that are useless for user identification

Resulting cleaned dataset

CustomerId	Location	Product1	Product2	...	ProductN
43	London	45	0	...	1
15	Coventry	0	15	...	1
145	Liverpool	0	0	...	0
27	Durham	0	7	...	34
4	Leeds	1	5	...	2
...	...	...	...	...	...

Map Geographic Locations from IP Address.

Determine “similar” groups of customers by **clustering**

For a given customer recommend the most frequent items accessed by the customers from their group

Data collection

Data preprocessing

Feature extraction

Cleaning

Feature selection & transformation

Analytical processing